Vishwas Kothari

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Education

University of Colorado Boulder — Aug 2025 – May 2027
Professional Master's in Computer Science — GPA: 3.74/4.0 — *Boulder, CO*
Relevant Coursework: Neural Networks & Deep Learning, Theory of Machine Learning, Design & Analysis of Algorithms, Database Systems, Linear Programming

Vellore Institute of Technology — Aug 2021 – Jun 2025
B.Tech in Information Technology — GPA: 3.4/4.0 — *Vellore, India*

Research Experience

Space Applications Centre – Indian Space Research Organisation (ISRO) — *Ahmedabad, Gujarat*
Research Intern, Biological Oceanography Division — Jan 2025 – Jul 2025

- Built end-to-end Python pipelines (SciPy, NumPy, Pandas, xarray) for data collection, preprocessing, and management across multi-source satellite datasets (NASA NOMAD, MODIS Aqua, OCM-3).
- Implemented a Levenberg–Marquardt optimization framework for bio-optical parameter retrieval from remote sensing reflectance, achieving $R^2 = 0.987$ on reconstruction ($n = 105$) and log-space $R^2 = 0.879$ on validation against in-situ reference data.
- Developed an inter-sensor colocation and validation pipeline (reprojection, resampling, pixel-wise merging) with RMSE, MAE, R^2, and bias metrics and 6 visualization types; documented procedures in a 60+ page technical report and presented updates to research staff across divisions.

Projects

XAI Credit Lens — Python, Scikit-learn, XGBoost, SHAP, LIME, DiCE, Optuna, Streamlit — GitHub — 2026

- Trained, validated, and evaluated tree-based ML models (XGBoost, LightGBM, Random Forest; Optuna-tuned over 100 trials, Stratified 5-Fold CV) achieving 0.782 test AUC on a 30K-record dataset.
- Integrated SHAP, LIME, and DiCE counterfactual explanations to make model predictions interpretable to non-technical stakeholders, mitigating over-reliance on opaque model outputs.
- Built an automated fairness audit across 4 protected attributes (zero violations) and deployed an interactive Streamlit dashboard with end-user-facing decision insights.

FinAgent-Eval — Python, PyTorch, Vision-Language Models, LoRA, GitHub Actions CI — GitHub — 2026

- Built a reproducible evaluation harness benchmarking 5 ML models on 397 QA pairs across 4 categories, with grader-level normalization and 95% bootstrap confidence intervals on all reported scores.
- Designed a failure-mode taxonomy and paraphrase reliability probe that surfaced model collapse on numerical reasoning (0.11 ANLS) and 0/6 paraphrase consistency, informing data-driven model-selection recommendations.
- Shipped 12 passing unit tests via GitHub Actions CI and a live evaluation dashboard, with full documentation and a reproducible Colab notebook for stakeholder review.

Financial Document VQA — PyTorch, LayoutLMv3, Donut, Pix2Struct, LoRA — GitHub — 2026

- Curated a 397-question benchmark (79 images, 10 companies, 4 categories) and benchmarked 4 vision-language architectures, identifying a 63–83% performance drop from generic benchmarks and a ranking reversal on dense tables.
- Conducted structured error analysis (structural errors 65%) and applied LoRA fine-tuning for a +22% ANLS gain on a held-out test split, versus negative transfer from generic fine-tuning data.

Implicit Bias of Adam vs. SGD — PyTorch, CVXPY, NumPy — GitHub — 2026

- Verified margin convergence via convex programming across 5 seeds: GD $\rightarrow \ell_2$ max-margin (cosine 0.990), Adam $\rightarrow \ell_\infty$ max-margin (cosine 0.988).
- Demonstrated SGD's simplicity bias causes a 30% out-of-distribution accuracy drop versus Adam's 13% on an MNIST spurious-correlation benchmark.

Technical Skills

Languages & Data: Python, SQL, Java, C/C++, NumPy, Pandas, SciPy
ML & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, SHAP, LIME, LoRA Fine-Tuning, statistical analysis, data visualization (Matplotlib, Seaborn, Plotly)
Tools: Git, GitHub Actions, Linux, Jupyter, VS Code, Streamlit, Gradio, LaTeX

Publications, Certifications & Service

Publication: Kothari, V., Krishwanth, B. (2025). Empowering Survivors: Ethical AI for Countering Violence Against Women. *Springer.*
Certification: AWS Certified Cloud Practitioner (970/1000, 2024).
Peer Reviewer: *Public Health* and *Social Sciences and Humanities Open* (Elsevier Journals), 2026 – Present.